SCIENCE

Condensation of Determinants

Lewis Carroll

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[Pg 1]

IV. “Condensation of Determinants, Being a New and Brief Method for Computing their Arithmetical Values.”

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By the
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Communicated by the Rev. Bartholomew Price, M.A., F.R.S. Received May 15, 1866.

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If it be proposed to solve a set of simultaneous linear equations, not being all homogeneous, involving unknowns, or to test their compatibility when all are homogeneous, by the method of determinants, in these, as well as in other cases of common occurrence, it is necessary to compute the arithmetical values of one or more determinants—such, for example, as

comma 2nd Column 3 comma 3rd Column negative 2 2nd Row 1st Column 2 comma 2nd Column 1 comma 3rd Column 4 3rd Row 1st Column 3 comma 2nd Column 5 comma 3rd Column negative 1 EndDeterminant period" data-tex="\[\left|\begin{array}{rrr} 1, &3, &-2\\ 2, &1, & 4\\ 3, &5, &-1 \end{array}\right|. \]" src="images/2.svg" style="vertical-align: -4.016ex; width: 15.712ex; height: 9.163ex;" id="img_images_2.svg">

Now the only method, so far as I am aware, that has been hitherto employed for such a purpose, is that of multiplying each term of the first row or column by the determinant of its complemental minor, and affecting the products with the signs <img alt="plus" data-tex="\({+}\)" src="images/3.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_3.svg"> and <img alt="minus" data-tex="\({-}\)" src="images/4.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_4.svg"> alternately, the determinants required in the process being, in their turn, broken up in the same manner until determinants are finally arrived at sufficiently small for mental computation.

This process, in the above instance, would run thus:— <img alt="StartLayout 1st Row Start 3 By 3 Determinant 1st Row 1st Column 1 comma 2nd Column 3 comma 3rd Column negative 2 2nd Row 1st Column 2 comma 2nd Column 1 comma 3rd Column 4 3rd Row 1st Column 3 comma 2nd Column 5 comma 3rd Column negative 1 EndDeterminant equals 1 times Start 2 By 2 Determinant 1st Row 1st Column 1 comma 2nd Column 4 2nd Row 1st Column 5 comma 2nd Column negative 1 EndDeterminant minus 2 times Start 2 By 2 Determinant 1st Row 1st Column 3 comma 2nd Column negative 2 2nd Row 1st Column 5 comma 2nd Column negative 1 EndDeterminant plus 3 times Start 2 By 2 Determinant 1st Row 1st Column 3 comma 2nd Column negative 2 2nd Row 1st Column 1 comma 2nd Column 4 EndDeterminant 2nd Row equals negative 21 minus 14 plus 42 equals 7 period EndLayout" data-tex="\[\begin{array}{r} \left|\begin{array}{rrr} 1, & 3, & -2 \\ 2, & 1, & 4 \\ 3, & 5, & -1 \end{array}\right|=1 \times\left|\begin{array}{rr} 1, & 4 \\ 5, & -1 \end{array}\right|-2 \times\left|\begin{array}{ll} 3, & -2 \\ 5, & -1 \end{array}\right|+3 \times\left|\begin{array}{rr} 3, & -2 \\ 1, & 4 \end{array}\right| \\ =-21-14+42=7 . \end{array} \]" src="images/5.svg" style="vertical-align: -5.882ex; width: 69.257ex; height: 12.896ex;" id="img_images_5.svg">

But such a process, when the block consists of <img alt="16" data-tex="\({16}\)" src="images/6.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_6.svg">, <img alt="25" data-tex="\({25}\)" src="images/7.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_7.svg">, or more terms, is so tedious that the old method of elimination is much to be preferred for solving simultaneous equations; so that the new method, excepting for equations containing <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;"

id="img_images_8.svg"> or <img alt="3" data-tex="\({3}\)" src="images/9.svg" style="vertical-align: -0.05ex; width: 1.131ex; height: 1.557ex;" id="img_images_9.svg"> unknowns, is practically useless.

The new method of computation, which I now proceed to explain, and for which “Condensation” appears to be an appropriate name, will be found, I believe, to be far shorter and simpler than any hitherto employed.

In the following remarks I shall use the word “Block” to denote any number of terms arranged in rows and columns, and “interior of a block” to denote the block which remains when the first and last rows and columns are erased.

The process of “Condensation” is exhibited in the following rules, in which the given block is supposed to consist of <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_2"> rows and <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_3"> columns:—

(1) Arrange the given block, if necessary, so that no ciphers occur in its interior. This may be done either by transposing rows or columns, or by adding to certain rows the several terms of other rows multiplied by certain multipliers.

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(2) Compute the determinant of every minor consisting of four adjacent terms. These values will constitute a second block, consisting of <img alt="left parenthesis n minus 1 right parenthesis" data-tex="\({(n - 1)}\)" src="images/10.svg" style="vertical-align: -0.561ex; width: 7.015ex; height: 2.253ex;" id="img_images_10.svg"> rows and <img alt="left parenthesis n minus 1 right parenthesis" data-tex="\({(n - 1)}\)" src="images/10.svg" style="vertical-align: -0.561ex; width: 7.015ex; height: 2.253ex;" id="img_images_10.svg_1"> columns.

(3) Condense this second block in the same manner, dividing each term, when found, by the corresponding term in the interior of the first block.

(4) Repeat this process as often as may be necessary (observing that in condensing any block of the series, the <img alt="r" data-tex="\({r}\)" src="images/11.svg" style="vertical-align: -0.025ex; width: 1.02ex; height: 1.025ex;" id="img_images_11.svg">th for example, the terms so found must be divided by the corresponding terms in the interior of the <img alt="left parenthesis r minus 1 right parenthesis" data-tex="\({(r - 1)}\)" src="images/12.svg" style="vertical-align: -0.561ex; width: 6.677ex; height: 2.253ex;" id="img_images_12.svg">th block), until the block is condensed to a single term, which will be the required value.

As an instance of the foregoing rules, let us take the block <img alt="Start 4 By 4 Determinant 1st Row 1st Column negative 2 2nd Column negative 1 3rd Column negative 1 4th Column negative 4 2nd Row 1st Column negative 1 2nd Column negative 2 3rd Column negative 1 4th Column negative 6 3rd Row 1st Column negative 1 2nd Column negative 1 3rd

Column 2 4th Column 4 4th Row 1st Column 2 2nd Column 1 3rd Column negative 3 4th Column negative 8 EndDeterminant period" data-tex="\[\left|\begin{array}{rrrr} -2 &-1 &-1 &-4\\ -1 &-2 &-1 &-6\\ -1 &-1 & 2 & 4\\ 2 & 1 &-3 &-8 \end{array}\right|. \]" src="images/13.svg" style="vertical-align: -5.6ex; width: 23.128ex; height: 12.33ex;" id="img_images_13.svg">

By rule (2) this is condensed into <img alt="Start 3 By 3 Determinant 1st Row 1st Column 3 2nd Column negative 1 3rd Column 2 2nd Row 1st Column negative 1 2nd Column negative 5 3rd Column 8 3rd Row 1st Column 1 2nd Column 1 3rd Column negative 4 EndDeterminant" data-tex="\({\left|\begin{array}{rrr} 3 &-1 & 2\\ -1 &-5 & 8\\ 1 & 1 &-4 \end{array}\right| }\)" src="images/14.svg" style="vertical-align: -4.016ex; width: 16.968ex; height: 9.163ex;" id="img_images_14.svg">; this, again, by rule (3), is condensed into <img alt="Start 2 By 2 Determinant 1st Row 1st Column 8 2nd Column negative 2 2nd Row 1st Column negative 4 2nd Column 6 EndDeterminant" data-tex="\({\left|\begin{array}{rr} 8 &-2\\ -4 & 6 \end{array}\right|}\)" src="images/15.svg" style="vertical-align: -2.432ex; width: 11.814ex; height: 5.995ex;" id="img_images_15.svg">; and this, by rule (4), into <img alt="negative 8" data-tex="\({-8}\)" src="images/16.svg" style="vertical-align: -0.188ex; width: 2.891ex; height: 1.695ex;" id="img_images_16.svg">, which is the required value.

The simplest method of working this rule appears to be to arrange the series of blocks one under another, as here exhibited; it will then be found very easy to pick out the divisors required in rules (3) and (4). <img alt="Start 4 By 4 Determinant 1st Row 1st Column negative 2 2nd Column negative 1 3rd Column negative 1 4th Column negative 4 2nd Row 1st Column negative 1 2nd Column negative 2 3rd Column negative 1 4th Column negative 6 3rd Row 1st Column negative 1 2nd Column negative 1 3rd Column 2 4th Column 4 4th Row 1st Column 2 2nd Column 1 3rd Column negative 3 4th Column negative 8 EndDeterminant" data-tex="\[\left|\begin{array}{rrrr} -2 &-1 &-1 &-4\\ -1 &-2 &-1 &-6\\ -1 &-1 & 2 & 4\\ 2 & 1 &-3 &-8 \end{array}\right| \]" src="images/17.svg" style="vertical-align: -5.6ex; width: 22.122ex; height: 12.33ex;" id="img_images_17.svg"> <img alt="Start 3 By 3 Determinant 1st Row 1st Column 3 2nd Column negative 1 3rd Column 2 2nd Row 1st Column negative 1 2nd Column negative 5 3rd Column 8 3rd Row 1st Column 1 2nd Column 1 3rd Column negative 4 EndDeterminant" data-tex="\[\left|\begin{array}{rrr} 3 &-1 &2\\ -1 &-5 &8\\ 1 & 1 &-4 \end{array}\right| \]" src="images/18.svg" style="vertical-align: -4.016ex; width: 16.968ex; height: 9.163ex;" id="img_images_18.svg"> <img alt="Start 2 By 2 Determinant 1st Row 1st Column 8 2nd Column negative 2 2nd Row 1st Column negative 4 2nd Column 6 EndDeterminant" data-tex="\[\left|\begin{array}{rr} 8 &-2\\ -4 & 6 \end{array}\right| \]" src="images/19.svg" style="vertical-align: -2.432ex; width: 11.814ex; height: 5.995ex;"

id="img_images_19.svg"> <img alt="negative 8 period" data-tex="\[-8. \]" src="images/20.svg" style="vertical-align: -0.188ex; width: 3.52ex; height: 1.695ex;" id="img_images_20.svg">

This process cannot be continued when ciphers occur in the interior of any one of the blocks, since infinite values would be introduced by employing them as divisors. When they occur in the given block itself, it may be rearranged as has been already mentioned; but this cannot be done when they occur in any one of the derived blocks; in such a case the given block must be rearranged as circumstances require, and the operation commenced anew.

The best way of doing this is as follows:—

Suppose a cipher to occur in the <img alt="h" data-tex="\({h}\)" src="images/21.svg" style="vertical-align: -0.025ex; width: 1.303ex; height: 1.595ex;" id="img_images_21.svg">th row and <img alt="k" data-tex="\({k}\)" src="images/22.svg" style="vertical-align: -0.025ex; width: 1.179ex; height: 1.595ex;" id="img_images_22.svg">th column of one of the derived blocks (reckoning[Pg 3] both row and column from the nearest corner of the block); find the term in the <img alt="h" data-tex="\({h}\)" src="images/21.svg" style="vertical-align: -0.025ex; width: 1.303ex; height: 1.595ex;" id="img_images_21.svg_1">th row and <img alt="k" data-tex="\({k}\)" src="images/22.svg" style="vertical-align: -0.025ex; width: 1.179ex; height: 1.595ex;" id="img_images_22.svg_1">th column of the given block (reckoning from the corresponding corner), and transpose rows or columns cyclically until it is left in an outside row or column. When the necessary alterations have been made in the derived blocks, it will be found that the cipher now occurs in an outside row or column, and therefore need no longer be used as a divisor.

The advantage of cyclical transposition is, that most of the terms in the new blocks will have been computed already, and need only be copied; in no case will it be necessary to compute more than one new row or column for each block of the series.

In the following instance it will be seen that in the first series of blocks a cipher occurs in the interior of the third. We therefore abandon the process at that point and begin again, rearranging the given block by transferring the top row to the bottom; and the cipher, when it occurs, is now found in an exterior row. It will be observed that in each block of the new series, there is only one new row to be computed; the other rows are simply copied from the work already done. <img alt="StartLayout 1st Row 1st Column Start 5 By 5 Determinant 1st Row 1st Column 2 2nd Column negative 1 3rd Column 2 4th Column 1 5th Column negative 3 2nd Row 1st Column 1 2nd Column 2 3rd Column 1 4th Column negative 1 5th Column 2 3rd Row 1st Column 1 2nd Column negative 1 3rd Column negative 2 4th Column negative 1 5th Column negative 1 4th Row 1st Column 2 2nd Column 1 3rd Column negative 1 4th Column negative 2 5th Column negative 1 5th Row 1st Column 1 2nd Column negative 2 3rd Column negative 1 4th Column negative 1 5th Column 2 EndDeterminant 2nd Column Start 5 By 5

Determinant 1st Row 1st Column 1 2nd Column 2 3rd Column 1 4th Column negative 1 5th Column 2 2nd Row 1st Column 1 2nd Column negative 1 3rd Column negative 2 4th Column negative 1 5th Column negative 1 3rd Row 1st Column 2 2nd Column 1 3rd Column negative 1 4th Column negative 2 5th Column negative 1 4th Row 1st Column 1 2nd Column negative 2 3rd Column negative 1 4th Column negative 1 5th Column 2 5th Row 1st Column 2 2nd Column negative 1 3rd Column 2 4th Column 1 5th Column negative 3 EndDeterminant 3rd Column Start 4 By 4 Determinant 1st Row 1st Column negative 3 2nd Column negative 3 3rd Column negative 3 4th Column 3 2nd Row 1st Column 3 2nd Column 3 3rd Column 3 4th Column negative 1 3rd Row 1st Column negative 5 2nd Column negative 3 3rd Column negative 1 4th Column negative 5 4th Row 1st Column 3 2nd Column negative 5 3rd Column 1 4th Column 1 EndDeterminant 2nd Row 1st Column Start 4 By 4 Determinant 1st Row 1st Column 5 2nd Column negative 5 3rd Column negative 3 4th Column negative 1 2nd Row 1st Column negative 3 2nd Column negative 3 3rd Column negative 3 4th Column 3 3rd Row 1st Column 3 2nd Column 3 3rd Column 3 4th Column negative 1 4th Row 1st Column negative 5 2nd Column negative 3 3rd Column negative 1 4th Column negative 5 EndDeterminant 2nd Column Start 3 By 3 Determinant 1st Row 1st Column 0 2nd Column 0 3rd Column 6 2nd Row 1st Column 6 2nd Column negative 6 3rd Column 8 3rd Row 1st Column negative 17 2nd Column 8 3rd Column negative 4 EndDeterminant 3rd Row 1st Column Start 3 By 3 Determinant 1st Row 1st Column negative 15 2nd Column 6 3rd Column 12 2nd Row 1st Column 0 2nd Column 0 3rd Column 6 3rd Row 1st Column 6 2nd Column negative 6 3rd Column 8 EndDeterminant 2nd Column Start 2 By 2 Determinant 1st Row 0 12 2nd Row 18 40 EndDeterminant 4th Row 1st Column Blank 2nd Column 36 period EndLayout" data-tex="\[\begin{array}{rlr} \left|\begin{array}{rrrr} 2 & -1 & 2 & 1 & -3 \\ 1 & 2 & 1 & -1 & 2 \\ 1 & -1 & -2 & -1 & -1 \\ 2 & 1 & -1 & -2 & -1 \\ 1 & -2 & -1 & -1 & 2 \end{array}\right| & \left|\begin{array}{rrrr} 1 & 2 & 1 & -1 & 2 \\ 1 & -1 & -2 & -1 & -1 \\ 2 & 1 & -1 & -2 & -1 \\ 1 & -2 & -1 & -1 & 2 \\ 2 & -1 & 2 & 1 & -3 \end{array}\right| & \left|\begin{array}{rrrr} -3 & -3 & -3 & 3 \\ 3 & 3 & 3 & -1 \\ -5 & -3 & -1 & -5 \\ 3 & -5 & 1 & 1 \end{array}\right| \\ \left|\begin{array}{rrrr} 5 & -5 & -3 & -1 \\ -3 & -3 & -3 & 3 \\ 3 & 3 & 3 & -1 \\ -5 & -3 & -1 & -5 \end{array}\right| & \left|\begin{array}{rrr} 0 & 0 & 6 \\ 6 & -6 & 8 \\ -17 & 8 & -4 \end{array}\right| \\ \left|\begin{array}{rrr} -15 & 6 & 12 \\ 0 & 0 & 6 \\ 6 & -6 & 8 \end{array}\right| & \left|\begin{array}{rr} 0 & 12 \\ 18 & 40 \end{array}\right| \\ & 36 . \end{array} \]" src="images/23.svg" style="vertical-align: -20.701ex; width: 79.941ex; height: 42.534ex;" id="img_images_23.svg">

The fact that, whenever ciphers occur in the interior of a derived block, it is necessary to recommence the operation, may be thought a great obstacle to the use of this method; but I believe it will be found in practice that, even though this should occur several times in the course of one operation, the whole amount of labour will

still be much less than that involved in the old process of computation.

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I now proceed to give a proof of the validity of this process, deduced from a well-known theorem in determinants; and in doing so, I shall use the word “adjugate” in the following sense:—if there be a square block, and if a new block be formed, such that each of its terms is the determinant of the complemental minor of the corresponding term of the first block, the second block is said to be adjugate to the first.

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The theorem referred to is the following:—

“If the determinant of a block <img alt="equals upper R" data-tex="\({= R}\)" src="images/24.svg" style="vertical-align: -0.05ex; width: 4.106ex; height: 1.595ex;" id="img_images_24.svg">, the determinant of any minor of the <img alt="m" data-tex="\({m}\)" src="images/25.svg" style="vertical-align: -0.025ex; width: 1.986ex; height: 1.025ex;" id="img_images_25.svg">th degree of the adjugate block is the product of <img alt="upper R Superscript m minus 1" data-tex="\({R^{{}m - 1}}\)" src="images/26.svg" style="vertical-align: -0.05ex; width: 5.354ex; height: 1.937ex;" id="img_images_26.svg"> and the coefficient which, in <img alt="upper R" data-tex="\({R}\)" src="images/27.svg" style="vertical-align: -0.05ex; width: 1.717ex; height: 1.595ex;" id="img_images_27.svg">, multiplies the determinant of the corresponding minor.”

Let us first take a block of <img alt="9" data-tex="\({9}\)" src="images/28.svg" style="vertical-align: -0.05ex; width: 1.131ex; height: 1.557ex;" id="img_images_28.svg"> terms, <img alt="Start 3 By 3 Determinant 1st Row 1st Column a Subscript 1 comma 1 Baseline 2nd Column a Subscript 1 comma 2 Baseline 3rd Column a Subscript 1 comma 3 Baseline 2nd Row 1st Column a Subscript 2 comma 1 Baseline 2nd Column a Subscript 2 comma 2 Baseline 3rd Column a Subscript 2 comma 3 Baseline 3rd Row 1st Column a Subscript 3 comma 1 Baseline 2nd Column a Subscript 3 comma 2 Baseline 3rd Column a Subscript 3 comma 3 Baseline EndDeterminant equals upper R semicolon" data-tex="\[\left|\begin{array}{ccc} a_{1,1} &a_{1,2} &a_{1,3}\\ a_{2,1} &a_{2,2} &a_{2,3}\\ a_{3,1} &a_{3,2} &a_{3,3} \end{array}\right| = R; \]" src="images/29.svg" style="vertical-align: -4.14ex; width: 23.945ex; height: 9.41ex;" id="img_images_29.svg"> and let <img alt="alpha Subscript 1 comma 1" data-tex="\({\alpha_{1,1}}\)" src="images/30.svg" style="vertical-align: -0.648ex; width: 3.68ex; height: 1.648ex;" id="img_images_30.svg"> represent the determinant of the complemental minor of <img alt="a Subscript 1 comma 1" data-tex="\({a_{1,1}}\)" src="images/31.svg" style="vertical-align: -0.648ex; width: 3.429ex; height: 1.648ex;" id="img_images_31.svg">, and so on.

If we “condense” this, by the method already given, we get the block <img alt="Start 2 By 2 Matrix 1st Row 1st Column alpha Subscript 3 comma 3 Baseline

2nd Column alpha Subscript 3 comma 1 Baseline 2nd Row 1st Column alpha Subscript 1 comma 3 Baseline 2nd Column alpha Subscript 1 comma 1 Baseline EndMatrix comma" data-tex="\({\left\{\begin{array}{cc} \alpha_{3,3} &\alpha_{3,1}\\ \alpha_{1,3} &\alpha_{1,1} \end{array}\right\},}\)" src="images/32.svg" style="vertical-align: -2.828ex; width: 16.973ex; height: 6.787ex;" id="img_images_32.svg"> and, by the theorem above cited, the determinant of this, viz. <img alt="Start 2 By 2 Determinant 1st Row 1st Column alpha Subscript 3 comma 3 Baseline 2nd Column alpha Subscript 3 comma 1 Baseline 2nd Row 1st Column alpha Subscript 1 comma 3 Baseline 2nd Column alpha Subscript 1 comma 1 Baseline EndDeterminant equals upper R times a Subscript 2 comma 2 Baseline period" data-tex="\[\left|\begin{array}{cc} \alpha_{3,3} &\alpha_{3,1}\\ \alpha_{1,3} &\alpha_{1,1} \end{array}\right| = R \times a_{2,2}. \]" src="images/33.svg" style="vertical-align: -2.515ex; width: 24.95ex; height: 6.161ex;" id="img_images_33.svg">

Hence <img alt="upper R equals StartFraction Start 2 By 2 Determinant 1st Row 1st Column alpha Subscript 3 comma 3 Baseline 2nd Column alpha Subscript 3 comma 1 Baseline 2nd Row 1st Column alpha Subscript 1 comma 3 Baseline 2nd Column alpha Subscript 1 comma 1 Baseline EndDeterminant Over a Subscript 2 comma 2 Baseline EndFraction comma" data-tex="\[R=\frac{\left|\begin{array}{ll} \alpha_{3,3} & \alpha_{3,1} \\ \alpha_{1,3} & \alpha_{1,1} \end{array}\right|}{a_{2,2}}, \]" src="images/34.svg" style="vertical-align: -2.2ex; width: 19.751ex; height: 9.401ex;" id="img_images_34.svg"> which proves the rule.

Secondly, let us take a block of <img alt="16" data-tex="\({16}\)" src="images/6.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_6.svg_1"> terms: <img alt="Start 3 By 3 Determinant 1st Row 1st Column a Subscript 1 comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript 1 comma 4 Baseline 2nd Row 1st Column vertical ellipsis 2nd Column Blank 3rd Column vertical ellipsis 3rd Row 1st Column a Subscript 4 comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript 4 comma 4 Baseline EndDeterminant equals upper R period" data-tex="\[\left|\begin{array}{ccc} a_{1,1}&\dots &a_{1,4}\\ \vdots & &\vdots \\ a_{4,1}&\dots &a_{4,4} \end{array}\right| = R. \]" src="images/35.svg" style="vertical-align: -4.098ex; width: 22.409ex; height: 9.328ex;" id="img_images_35.svg"> If we “condense” this, we get a block of <img alt="9" data-tex="\({9}\)" src="images/28.svg" style="vertical-align: -0.05ex; width: 1.131ex; height: 1.557ex;" id="img_images_28.svg_1"> terms; let us denote it by <img alt="Start 3 By 3 Matrix 1st Row 1st Column b Subscript 1 comma 1 Baseline 2nd Column ellipsis 3rd Column b Subscript 1 comma 3 Baseline 2nd Row 1st Column vertical ellipsis 2nd Column Blank 3rd Column vertical ellipsis 3rd Row 1st Column b Subscript 3 comma 1 Baseline 2nd Column ellipsis 3rd Column b Subscript 3 comma 3 Baseline EndMatrix comma in which b Subscript 1 comma 1 Baseline equals Start 2 By 2 Determinant 1st Row 1st Column a Subscript 1 comma 1 Baseline 2nd Column a Subscript 1 comma 2 Baseline 2nd Row

1st Column a Subscript 2 comma 1 Baseline 2nd Column a Subscript 2 comma 2 Baseline EndDeterminant comma ampersand c period" data-tex="\[\left\{\begin{array}{ccc} b_{1,1} &\dots &b_{1,3}\\ \vdots & &\vdots \\ b_{3,1} &\dots &b_{3,3} \end{array}\right\}, \text{ in which } b_{1,1} = \left|\begin{array}{cc} a_{1,1} &a_{1,2}\\ a_{2,1} &a_{2,2} \end{array}\right|, \text{ \&c.} \]" src="images/36.svg" style="vertical-align: -4.098ex; width: 55.112ex; height: 9.328ex;" id="img_images_36.svg">

If we “condense” this block again, we get a block of <img alt="4" data-tex="\({4}\)" src="images/37.svg" style="vertical-align: 0; width: 1.131ex; height: 1.532ex;" id="img_images_37.svg"> terms, each of which, by the preceding paragraph, is the determinant of <img alt="9" data-tex="\({9}\)" src="images/28.svg" style="vertical-align: -0.05ex; width: 1.131ex; height: 1.557ex;" id="img_images_28.svg_2"> terms of the original block; that is to say, we get the block <img alt="Start 2 By 2 Matrix 1st Row 1st Column alpha Subscript 4 comma 4 2nd Column alpha Subscript 4 comma 1 2nd Row 1st Column alpha Subscript 1 comma 4 2nd Column alpha Subscript 1 comma 1 EndMatrix" data-tex="\({\left\{\begin{array}{cc} \alpha_{4,4} &\alpha_{4,1}\\ \alpha_{1,4} &\alpha_{1,1} \end{array}\right\}}\)" src="images/38.svg" style="vertical-align: -2.828ex; width: 15.967ex; height: 6.787ex;" id="img_images_38.svg">; but, by the theorem already quoted, <img alt="Start 2 By 2 Determinant 1st Row 1st Column alpha Subscript 4 comma 4 Baseline 2nd Column alpha Subscript 4 comma 1 Baseline 2nd Row 1st Column alpha Subscript 1 comma 4 Baseline 2nd Column alpha Subscript 1 comma 1 Baseline EndDeterminant equals upper R times b Subscript 2 comma 2" data-tex="\({\left|\begin{array}{cc} \alpha_{4,4} &\alpha_{4,1}\\ \alpha_{1,4} &\alpha_{1,1} \end{array}\right| = R \times b_{2,2}}\)" src="images/39.svg" style="vertical-align: -2.515ex; width: 24.095ex; height: 6.161ex;" id="img_images_39.svg">; therefore <img alt="upper R equals StartFraction Start 2 By 2 Determinant 1st Row 1st Column alpha Subscript 4 comma 4 Baseline 2nd Column alpha Subscript 4 comma 1 Baseline 2nd Row 1st Column alpha Subscript 1 comma 4 Baseline 2nd Column alpha Subscript 1 comma 1 Baseline EndDeterminant Over b Subscript 2 comma 2 Baseline EndFraction" data-tex="\({R = \dfrac{\left|\begin{array}{cc} \alpha_{4,4} &\alpha_{4,1}\\ \alpha_{1,4} &\alpha_{1,1} \end{array}\right|}{b_{2,2}}}\)" src="images/40.svg" style="vertical-align: -2.2ex; width: 19.122ex; height: 9.401ex;" id="img_images_40.svg">; that is, <img alt="upper R" data-tex="\({R}\)" src="images/27.svg" style="vertical-align: -0.05ex; width: 1.717ex; height: 1.595ex;" id="img_images_27.svg_1"> may be obtained by “condensing” the block <img alt="Start 2 By 2 Matrix 1st Row 1st Column alpha Subscript 4 comma 4 2nd Column alpha Subscript 4 comma 1 2nd Row 1st Column alpha Subscript 1 comma 4 2nd Column alpha Subscript 1 comma 1 EndMatrix" data-tex="\({\left\{\begin{array}{cc} \alpha_{4,4} &\alpha_{4,1}\\ \alpha_{1,4} &\alpha_{1,1} \end{array}\right\}}\)" src="images/38.svg" style="vertical-align: -2.828ex; width: 15.967ex; height:

6.787ex;" id="img_images_38.svg_1">.

This proves the rule for a block of <img alt="16" data-tex="\({16}\)" src="images/6.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_6.svg_2"> terms; and similar proofs might be given for larger blocks.

[Pg 5]

<hr class="tb">

I shall conclude by showing how this process may be applied to the solution of simultaneous linear equations.

If we take a block consisting of <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_4"> rows and <img alt="left parenthesis n plus 1 right parenthesis" data-tex="\({(n + 1)}\)" src="images/41.svg" style="vertical-align: -0.561ex; width: 7.015ex; height: 2.253ex;" id="img_images_41.svg"> columns, and “condense” it, we reduce it at last to <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_1"> terms, the first of which is the determinant of the first <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_5"> columns, the other of the last <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_6"> columns.

Hence, if we take the <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_7"> simultaneous equations, <img alt="StartLayout 1st Row 1st Column Blank 2nd Column a Subscript 1 comma 1 Baseline x 1 plus a Subscript 1 comma 2 Baseline x 2 plus midline horizontal ellipsis midline horizontal ellipsis plus a Subscript 1 comma n Baseline x Subscript n plus a Subscript 1 comma n plus 1 3rd Column equals 0 comma 2nd Row 1st Column Blank 2nd Column midline horizontal ellipsis midline horizontal ellipsis midline horizontal ellipsis

-3.589ex; width: 46.711ex; height: 8.31ex;" id="img_images_42.svg"> and if we condense the whole block of coefficients and constants, viz. <img alt="Start 3 By 3 Matrix 1st Row 1st Column a Subscript 1 comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript 1 comma n plus 1 Baseline 2nd Row 1st Column vertical ellipsis 2nd Column Blank 3rd Column vertical ellipsis 3rd Row 1st Column a Subscript n comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript n comma n plus 1 Baseline EndMatrix comma" data-tex="\[\left\{\begin{array}{ccc} a_{1,1} &\dots &a_{1,n + 1}\\ \vdots & &\vdots \\ a_{n,1} &\dots &a_{n,n + 1} \end{array}\right\}, \]" src="images/43.svg" style="vertical-align: -4.098ex; width: 23.151ex; height: 9.328ex;" id="img_images_43.svg"> we reduce it at last to <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_2"> terms: let us denote them by <img alt="upper S" data-tex="\({S}\)" src="images/44.svg" style="vertical-align: -0.05ex; width: 1.459ex; height: 1.645ex;" id="img_images_44.svg">, <img alt="upper T" data-tex="\({T}\)" src="images/45.svg" style="vertical-align: 0; width: 1.593ex; height: 1.532ex;" id="img_images_45.svg">, so that <img alt="upper S equals Start 3 By 3 Determinant 1st Row 1st Column a Subscript 1 comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript 1 comma n Baseline 2nd Row 1st Column vertical ellipsis 2nd Column Blank 3rd Column vertical ellipsis 3rd Row 1st Column a Subscript n comma 1 Baseline 2nd Column ellipsis 3rd Column a Subscript n comma n Baseline EndDeterminant comma and upper T equals Start 3 By 3 Determinant 1st Row 1st Column a Subscript 1 comma 2 Baseline 2nd Column ellipsis 3rd Column a Subscript 1 comma n plus 1 Baseline 2nd Row 1st Column vertical ellipsis 2nd Column Blank 3rd Column vertical ellipsis 3rd Row 1st Column a Subscript n comma 2 Baseline 2nd Column ellipsis 3rd Column a Subscript n comma n plus 1 Baseline EndDeterminant period" data-tex="\[S = \left|\begin{array}{ccc} a_{1,1} &\dots &a_{1,n}\\ \vdots & &\vdots \\ a_{n,1} &\dots &a_{n,n} \end{array}\right|, \text{ and } T = \left|\begin{array}{ccc} a_{1,2} &\dots &a_{1,n+1}\\ \vdots & &\vdots \\ a_{n,2} &\dots &a_{n,n+1} \end{array}\right|. \]" src="images/46.svg" style="vertical-align: -4.098ex; width: 53.721ex; height: 9.328ex;" id="img_images_46.svg">

Now we know that <img alt="x 1 equals left parenthesis minus right parenthesis Superscript n Baseline StartFraction upper T Over upper S EndFraction" data-tex="\({x_{1} = (-)^{n} \dfrac{T}{S}}\)" src="images/47.svg" style="vertical-align: -1.602ex; width: 12.555ex; height: 4.663ex;" id="img_images_47.svg">, which may be written in the form <img alt="left parenthesis minus right parenthesis Superscript n Baseline upper S dot x 1 equals upper T" data-tex="\({(-)^{n}S \cdot x_{1} = T}\)" src="images/48.svg" style="vertical-align: -0.561ex; width: 14.653ex; height: 2.253ex;" id="img_images_48.svg">.

Hence the <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_3">

terms obtained by the process of condensation may be converted into an equation for <img alt="x 1" data-tex="\({x_{1}}\)" src="images/49.svg" style="vertical-align: -0.339ex; width: 2.282ex; height: 1.339ex;" id="img_images_49.svg">, by multiplying the first of them by <img alt="x 1" data-tex="\({x_{1}}\)" src="images/49.svg" style="vertical-align: -0.339ex; width: 2.282ex; height: 1.339ex;" id="img_images_49.svg_1">, affected with <img alt="plus" data-tex="\({+}\)" src="images/3.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_3.svg_1"> or <img alt="minus" data-tex="\({-}\)" src="images/4.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_4.svg_1">, according as <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_8"> is even or odd. The latter part of the rule may be simply expressed thus:—“place the signs <img alt="plus" data-tex="\({+}\)" src="images/3.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_3.svg_2"> and <img alt="minus" data-tex="\({-}\)" src="images/4.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_4.svg_2"> alternately over the several columns, beginning with the last, and the sign which occurs over the column containing <img alt="x 1" data-tex="\({x_{1}}\)" src="images/49.svg" style="vertical-align: -0.339ex; width: 2.282ex; height: 1.339ex;" id="img_images_49.svg_2"> is the sign with which <img alt="x 1" data-tex="\({x_{1}}\)" src="images/49.svg" style="vertical-align: -0.339ex; width: 2.282ex; height: 1.339ex;" id="img_images_49.svg_3"> is to be affected.”

When the value of <img alt="x 1" data-tex="\({x_{1}}\)" src="images/49.svg" style="vertical-align: -0.339ex; width: 2.282ex; height: 1.339ex;" id="img_images_49.svg_4"> has been thus found, it may be substituted in the first <img alt="left parenthesis n minus 1 right parenthesis" data-tex="\({(n - 1)}\)" src="images/10.svg" style="vertical-align: -0.561ex; width: 7.015ex; height: 2.253ex;" id="img_images_10.svg_2"> equations, and the same operation repeated on the new block, which will now consist of <img alt="left parenthesis n minus 1 right parenthesis" data-tex="\({(n - 1)}\)" src="images/10.svg" style="vertical-align: -0.561ex; width: 7.015ex; height: 2.253ex;" id="img_images_10.svg_3"> rows and <img alt="n" data-tex="\({n}\)" src="images/1.svg" style="vertical-align: -0.025ex; width: 1.357ex; height: 1.025ex;" id="img_images_1.svg_9"> columns. But in calculating the second series of blocks, it will be found that most of the work has been already done; in fact, of the <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_4"> determinants required in the new block, one has been already computed correctly, and the other so nearly so that it only requires the last column in each of the derived blocks to be corrected.

In the example given opposite, after writing <img alt="plus" data-tex="\({+}\)" src="images/3.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_3.svg_3"> and <img alt="minus" data-tex="\({-}\)" src="images/4.svg" style="vertical-align: -0.188ex; width: 1.76ex;

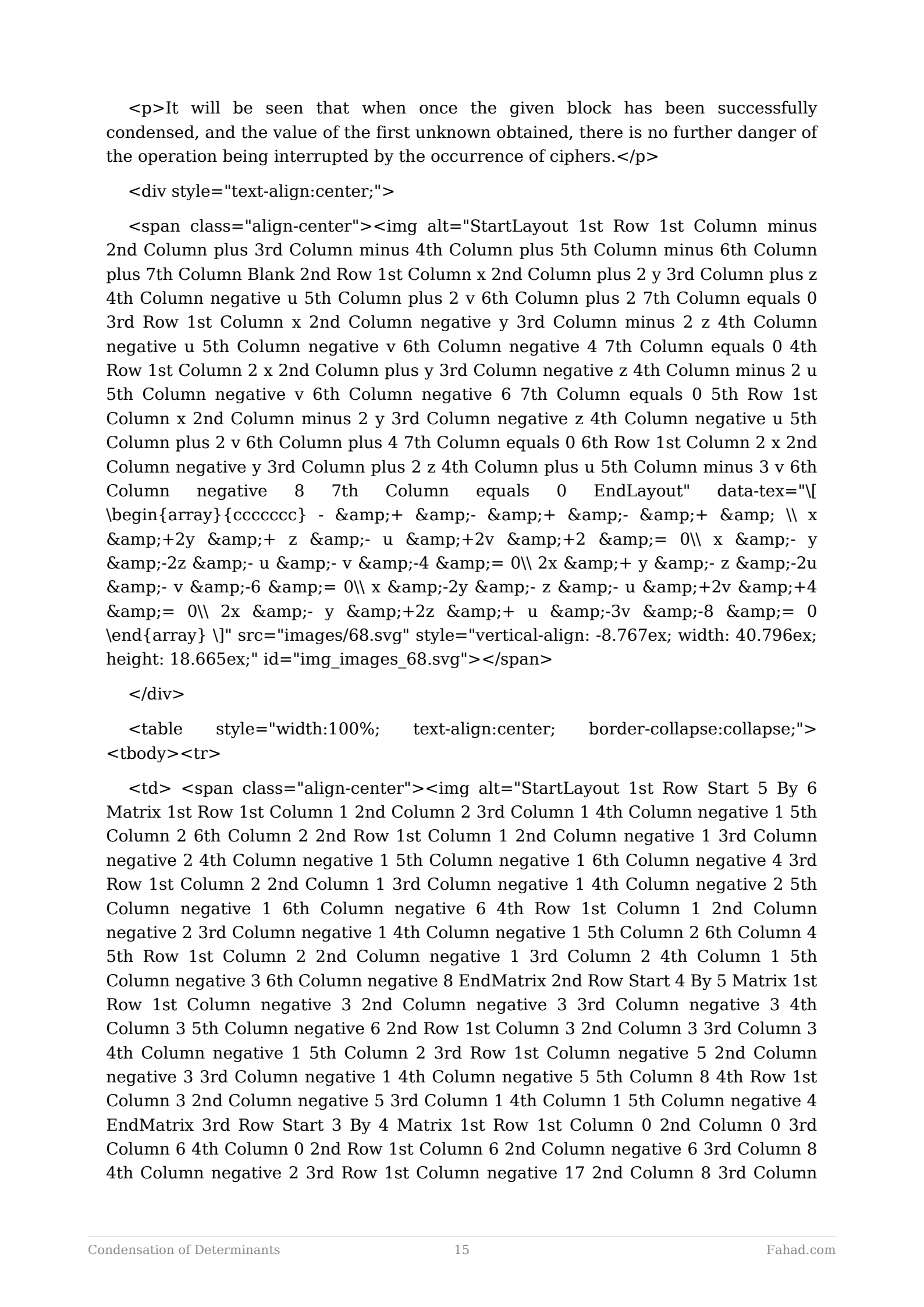
height: 1.507ex;" id="img_images_4.svg_3"> alternately over the columns, beginning with the last, we first condense the whole block, and thus obtain the <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_5"> terms <img alt="36" data-tex="\({36}\)" src="images/50.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_50.svg"> and <img alt="negative 72" data-tex="\({-72}\)" src="images/51.svg" style="vertical-align: -0.188ex; width: 4.023ex; height: 1.717ex;" id="img_images_51.svg">. Observing that the <img alt="x" data-tex="\({x}\)" src="images/52.svg" style="vertical-align: -0.025ex; width: 1.294ex; height: 1.025ex;" id="img_images_52.svg">-column has the sign <img alt="minus" data-tex="\({-}\)" src="images/4.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_4.svg_4"> placed over it, we multiply the <img alt="36" data-tex="\({36}\)" src="images/50.svg" style="vertical-align: -0.05ex; width: 2.262ex; height: 1.557ex;" id="img_images_50.svg_1"> by <img alt="negative x" data-tex="\({-x}\)" src="images/53.svg" style="vertical-align: -0.188ex; width: 3.054ex; height: 1.507ex;" id="img_images_53.svg">, and so form the equation <img alt="minus 36 x equals negative 72" data-tex="\({-36x = -72}\)" src="images/54.svg" style="vertical-align: -0.188ex; width: 12.356ex; height: 1.717ex;" id="img_images_54.svg">, which gives <img alt="x equals 2" data-tex="\({x = 2}\)" src="images/55.svg" style="vertical-align: -0.025ex; width: 5.442ex; height: 1.532ex;" id="img_images_55.svg">.

Hence the <img alt="x" data-tex="\({x}\)" src="images/52.svg" style="vertical-align: -0.025ex; width: 1.294ex; height: 1.025ex;" id="img_images_52.svg_1">-terms in the first four equations become respectively <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_6">, <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_7">, <img alt="4" data-tex="\({4}\)" src="images/37.svg" style="vertical-align: 0; width: 1.131ex; height: 1.532ex;" id="img_images_37.svg_1">, and <img alt="2" data-tex="\({2}\)" src="images/8.svg" style="vertical-align: 0; width: 1.131ex; height: 1.507ex;" id="img_images_8.svg_8">; adding these values to the constant terms in the same equations, we obtain a block of which we need only write down the last two columns, viz. <img alt="Start 4 By 2 Matrix 1st Row 1st Column 2 2nd Column 4 2nd Row 1st Column negative 1 2nd Column negative 2 3rd Row 1st Column negative 1 2nd Column negative 2 4th Row 1st Column 2 2nd Column 6 EndMatrix period" data-tex="\[\left|\begin{array}{rr} 2 & 4\\ -1 &-2\\ -1 &-2\\ 2 & 6 \end{array}\right|. \]" src="images/56.svg" style="vertical-align: -5.6ex; width: 12.821ex; height: 12.33ex;" id="img_images_56.svg">

[Pg 6]

We then condense these into the column<img alt="Start 3 By 1 Matrix 1st Row 0 2nd Row 0 3rd Row negative 2 EndMatrix" data-tex="\({\left|\begin{array}{r} 0\\ 0\\ -2 \end{array}\right|}\)" src="images/57.svg" style="vertical-align: -4.016ex; width: 6.661ex; height: 9.163ex;" id="img_images_57.svg">, and, supplying from the second block of the first series the column <img alt="Start 3 By 1 Matrix 1st Row 3 2nd Row negative 1 3rd Row negative 5 EndMatrix" data-tex="\({\left|\begin{array}{r} 3\\ -1\\ -5 \end{array}\right|}\)" src="images/58.svg" style="vertical-align: -4.016ex; width: 6.661ex; height: 9.163ex;" id="img_images_58.svg">, we obtain <img alt="Start 3 By 2 Matrix 1st Row 1st Column 3 2nd Column 0 2nd Row 1st Column negative 1 2nd Column 0 3rd Row 1st Column negative 5 2nd Column negative 2 EndMatrix" data-tex="\({\left|\begin{array}{rr} 3 &0\\ -1 &0\\ -5 & -2 \end{array}\right|}\)" src="images/59.svg" style="vertical-align: -4.016ex; width: 11.814ex; height: 9.163ex;" id="img_images_59.svg"> as the last two columns of the second block of the new series; and proceeding thus we ultimately obtain the two terms <img alt="12" data-tex="\({12}\)" src="images/60.svg" style="vertical-align: 0; width: 2.262ex; height: 1.507ex;" id="img_images_60.svg">, <img alt="12" data-tex="\({12}\)" src="images/60.svg" style="vertical-align: 0; width: 2.262ex; height: 1.507ex;" id="img_images_60.svg_1">. Observing that the <img alt="y" data-tex="\({y}\)" src="images/61.svg" style="vertical-align: -0.464ex; width: 1.109ex; height: 1.464ex;" id="img_images_61.svg">-column has the sign <img alt="plus" data-tex="\({+}\)" src="images/3.svg" style="vertical-align: -0.188ex; width: 1.76ex; height: 1.507ex;" id="img_images_3.svg_4"> placed over it, we multiply the first <img alt="12" data-tex="\({12}\)" src="images/60.svg" style="vertical-align: 0; width: 2.262ex; height: 1.507ex;" id="img_images_60.svg_2"> by <img alt="plus y" data-tex="\({+y}\)" src="images/62.svg" style="vertical-align: -0.464ex; width: 2.869ex; height: 1.783ex;" id="img_images_62.svg">, and so form the equation <img alt="12 y equals 12" data-tex="\({12y = 12}\)" src="images/63.svg" style="vertical-align: -0.464ex; width: 8.651ex; height: 1.971ex;" id="img_images_63.svg">, which gives <img alt="y equals 1" data-tex="\({y = 1}\)" src="images/64.svg" style="vertical-align: -0.464ex; width: 5.257ex; height: 1.971ex;" id="img_images_64.svg">. The values of <img alt="z" data-tex="\({z}\)" src="images/65.svg" style="vertical-align: -0.025ex; width: 1.057ex; height: 1.025ex;" id="img_images_65.svg">, <img alt="u" data-tex="\({u}\)" src="images/66.svg" style="vertical-align: -0.025ex; width: 1.294ex; height: 1.025ex;" id="img_images_66.svg">, and <img alt="v" data-tex="\({v}\)" src="images/67.svg" style="vertical-align: -0.025ex; width: 1.097ex; height: 1.025ex;" id="img_images_67.svg"> are similarly found.

It will be seen that when once the given block has been successfully condensed, and the value of the first unknown obtained, there is no further danger of the operation being interrupted by the occurrence of ciphers.



" data-tex="\[\begin{array}{ccccccc} - &+ &- &+ &- &+ & \\ x &+2y &+ z &- u &+2v &+2 &= 0\\ x &- y &-2z &- u &- v &-4 &= 0\\ 2x &+ y &- z &-2u &- v &-6 &= 0\\ x &-2y &- z &- u &+2v &+4 &= 0\\ 2x &- y &+2z &+ u &-3v &-8 &= 0 \end{array} \]" src="images/68.svg" style="vertical-align: -8.767ex; width: 40.796ex; height: 18.665ex;" id="img_images_68.svg">

<table style="width:100%; text-align:center; border-collapse:collapse;"> <tbody><tr>

<td>

negative 4 4th Column 6 EndMatrix 4th Row Start 2 By 3 Matrix 1st Row 1st Column 0 2nd Column 12 3rd Column 12 2nd Row 1st Column 18 2nd Column 40 3rd Column negative 8 EndMatrix 5th Row Start 1 By 2 Matrix 1st Row 1st Column 36 2nd Column negative 72 EndMatrix 6th Row therefore minus 36 x equals negative 72 7th Row x equals 2 EndLayout" data-tex="\[\begin{gathered} \left|\begin{array}{rrrrrr} 1 & 2 & 1 &-1 & 2 & 2\\ 1 &-1 &-2 &-1 &-1 &-4\\ 2 & 1 &-1 &-2 &-1 &-6\\ 1 &-2 &-1 &-1 & 2 & 4\\ 2 &-1 & 2 & 1 &-3 &-8 \end{array}\right| \\ \left|\begin{array}{rrrrr} -3 &-3 &-3 & 3 &-6\\ 3 & 3 & 3 &-1 & 2\\ -5 &-3 &-1 &-5 & 8\\ 3 &-5 & 1 & 1 &-4 \end{array}\right| \\ \left|\begin{array}{rrrr} 0 & 0 & 6 & 0\\ 6 &-6 & 8 &-2\\ -17 & 8 &-4 & 6 \end{array}\right| \\ \left|\begin{array}{rrr} 0 &12 &12\\ 18 &40 &-8 \end{array}\right| \\ \left|\begin{array}{rr} 36 & -72 \end{array}\right| \\ \therefore -36x = -72 \\ x = 2 \end{gathered} \]" src="images/69.svg" style="vertical-align: -26.64ex; width: 30.67ex; height: 54.412ex;" id="img_images_69.svg">

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<td>

<img alt="StartLayout 1st Row Start 4 By 2 Matrix 1st Row 1st Column 2 2nd Column 4 2nd Row 1st Column negative 1 2nd Column negative 2 3rd Row 1st Column negative 1 2nd Column negative 2 4th Row 1st Column 2 2nd Column 6 EndMatrix 2nd Row Start 3 By 2 Matrix 1st Row 1st Column 3 2nd Column 0 2nd Row 1st Column negative 1 2nd Column 0 3rd Row 1st Column negative 5 2nd Column negative 2 EndMatrix 3rd Row Start 2 By 2 Determinant 1st Row 1st Column 6 2nd Column 0 2nd Row 1st Column 8 2nd Column negative 2 EndDeterminant 4th Row Start 1 By 2 Matrix 1st Row 1st Column 12 2nd Column 12 EndMatrix 5th Row therefore 12 y equals 12 6th Row y equals 1 EndLayout" data-tex="\[\begin{gathered} \left|\begin{array}{rr} 2 & 4\\ -1 &-2\\ -1 &-2\\ 2 & 6 \end{array}\right| \\ \left|\begin{array}{rr} 3 & 0\\ -1 & 0\\ -5 &-2 \end{array}\right| \\ \left|\begin{array}{rr} 6 & 0\\ 8 &-2 \end{array}\right| \\ \left|\begin{array}{rr} 12 & 12 \end{array}\right| \\ \therefore 12y = 12 \\ y = 1 \end{gathered} \]" src="images/70.svg" style="vertical-align: -18.552ex; width: 11.814ex; height: 38.235ex;" id="img_images_70.svg">

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<td>

<img alt="StartLayout 1st Row Start 3 By 2 Matrix 1st Row 1st Column 2 2nd Column 6 2nd Row 1st Column negative 1 2nd Column negative 3 3rd Row 1st Column negative 1 2nd Column negative 1 EndMatrix 2nd Row Start 2 By 2 Determinant 1st Row 1st Column 3 2nd Column 0 2nd Row 1st Column negative 1 2nd Column negative 2 EndDeterminant 3rd Row Start 1 By 2

Matrix 1st Row 1st Column 6 2nd Column 6 EndMatrix 4th Row therefore minus 6 z equals 6 5th Row z equals negative 1 EndLayout" data-tex="\[\begin{gathered} \left|\begin{array}{rr} 2 & 6\\ -1 &-3\\ -1 &-1 \end{array}\right| \\ \left|\begin{array}{rr} 3 & 0\\ -1 &-2 \end{array}\right| \\ \left|\begin{array}{rr} 6 & 6 \end{array}\right| \\ \therefore -6z = 6 \\ z = -1 \end{gathered} \]" src="images/71.svg" style="vertical-align: -12.048ex; width: 11.814ex; height: 25.226ex;" id="img_images_71.svg">

</td>

<td>

<img alt="StartLayout 1st Row Start 2 By 2 Determinant 1st Row 1st Column 2 2nd Column 5 2nd Row 1st Column negative 1 2nd Column negative 1 EndDeterminant 2nd Row Start 1 By 2 Matrix 1st Row 1st Column 3 2nd Column 3 EndMatrix 3rd Row therefore 3 u equals 3 4th Row u equals 1 EndLayout" data-tex="\[\begin{gathered} \left|\begin{array}{rr} 2 & 5\\ -1 &-1 \end{array}\right| \\ \left|\begin{array}{rr} 3 & 3 \end{array}\right| \\ \therefore 3u = 3 \\ u = 1 \end{gathered} \]" src="images/72.svg" style="vertical-align: -7.127ex; width: 11.814ex; height: 15.385ex;" id="img_images_72.svg">

</td>

<td>

<img alt="StartLayout 1st Row Start 1 By 2 Matrix 1st Row 1st Column 2 2nd Column 4 EndMatrix 2nd Row therefore minus 2 v equals 4 3rd Row v equals negative 2 EndLayout" data-tex="\[\begin{gathered} \left|\begin{array}{rr} 2 & 4 \end{array}\right| \\ \therefore -2v = 4 \\ v = -2 \end{gathered} \]" src="images/73.svg" style="vertical-align: -3.79ex; width: 9.933ex; height: 8.71ex;" id="img_images_73.svg">

</td>

</tr> </tbody></table>

<hr class="tb">

<img alt="StartLayout 1st Row 1st Column minus 2nd Column plus 3rd Column minus 4th Column plus 5th Column Blank 2nd Row 1st Column 5 x 2nd Column plus 2 y 3rd Column minus 3 z 4th Column plus 3 5th Column equals 0 3rd Row 1st Column 3 x 2nd Column negative y 3rd Column minus 2 z 4th Column plus 7 5th Column equals 0 4th Row 1st Column 2 x 2nd Column plus 3 y 3rd Column plus z 4th Column negative 12 5th Column equals 0 EndLayout StartLayout 1st Row Start 3 By 4 Matrix 1st Row 1st Column 5 2nd Column 2 3rd Column negative 3 4th Column 3 2nd Row 1st Column 3 2nd Column negative 1 3rd Column negative 2 4th Column 7 3rd Row 1st Column 2 2nd Column 3 3rd Column 1 4th Column negative 12 EndMatrix 2nd Row Start 2 By 3 Matrix 1st Row 1st Column negative 11 2nd Column negative 7 3rd Column negative 15 2nd Row 1st Column 11 2nd Column 5 3rd Column 17 EndMatrix

3rd Row Start 1 By 2 Matrix 1st Row 1st Column negative 22 2nd Column 22 EndMatrix 4th Row therefore 22 x equals 22 5th Row x equals 1 EndLayout StartLayout 1st Row Start 2 By 2 Determinant 1st Row 1st Column negative 3 2nd Column 8 2nd Row 1st Column negative 2 2nd Column 10 EndDeterminant 2nd Row Start 1 By 2 Matrix 1st Row 1st Column negative 7 2nd Column negative 14 EndMatrix 3rd Row therefore minus 7 y equals negative 14 4th Row y equals 2 EndLayout StartLayout 1st Row Start 1 By 2 Matrix 1st Row 1st Column negative 3 2nd Column 12 EndMatrix 2nd Row therefore 3 z equals 12 3rd Row z equals 4 EndLayout" data-tex="\[\begin{array}{ccccc} - &+ &- &+ & \\ 5x &+2y &-3z &+ 3 &= 0\\ 3x &- y &-2z &+ 7 &= 0\\ 2x &+3y &+ z &-12 &= 0 \end{array}\\ \begin{gathered} \left|\begin{array}{rrrr} 5 & 2 &-3 & 3\\ 3 &-1 &-2 & 7\\ 2 & 3 & 1 &-12 \end{array}\right| \\ \left|\begin{array}{rrr} -11 &-7 &-15\\ 11 & 5 & 17 \end{array}\right| \\ \left|\begin{array}{cc} -22 & 22 \end{array}\right| \\ \therefore 22x =22 \\ x = 1 \end{gathered} \quad \begin{gathered} \vphantom{\begin{matrix}1\\1\\1\\1\end{matrix}} \left|\begin{array}{rr} -3 & 8\\ -2 &10 \end{array}\right| \\ \left|\begin{array}{cc} -7 & -14 \end{array}\right| \\ \therefore -7y = -14 \\ y = 2 \end{gathered} \quad \begin{gathered} \vphantom{\begin{matrix}1\\1\\1\\1\end{matrix}} \left|\begin{array}{rr} -3 & 12 \end{array}\right| \\ \therefore 3z = 12 \\ z = 4 \end{gathered} \]" src="images/74.svg" style="vertical-align: -31.278ex; width: 50.526ex; height: 38.009ex;" id="img_images_74.svg">

The Society then adjourned over the Whitsuntide Recess to Thursday, May 31.

<hr aria-hidden="true" class="chap x-ebookmaker-drop"> <h2 class="nobreak" id="TRANSCRIBERS_NOTE">TRANSCRIBER’S NOTE</h2>

Minor typographical corrections and presentational changes have been made without comment.

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<footer class="pg-boilerplate pgheader" id="pg-footer" lang="en">